

Supplementary Summer Examination – 2024

Course: B. Tech. Branch: Mechanical Engg./ Mech. Engg. Sandwich Sem: VII

Subject Code & Name: BTHM702 Industrial Engineering and Management

Max Marks: 60

Date: 03/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following		12
A) Enlist various forms of organization structures and describe any one of them with its advantages and disadvantages.	CO3	6
B) Explain the concept of scientific management by F.W. Taylor.	CO1	6
C) What is the significance of Industrial Engineering in productivity improvement?	CO1	6
Q. 2 Solve Any Two of the following		12
A) What are the roles of human resource management? Explain various challenges faced by an effective HRM.	CO2	6
B) What do you understand by job evolution? How it benefits to an industry?	CO3	6
C) What is performance appraisal? Explain the characteristics of a sound appraisal system.	CO1	6
Q. 3 Solve Any Two of the following		12
A) Distinguish between product and process layout with neat sketch.	CO2	6
B) What are the different types of manufacturing systems? Explain in detail with the help of neat sketch.	CO3	6
C) What do you understand by operations management? Explain the principles of operations management.	CO1	6
Q. 4 Solve Any Two of the following		12
A) What is line balancing? Explain the methods of line balancing.	CO2	6
B) Enlist and explain various steps involved in product design process with the help of a flow chart.	CO3	6
C) What is the significance of capacity planning in a manufacturing industry? Explain the various capacity planning activities in detail.	CO2	6
Q. 5 Solve Any Two of the following		12
A) Enlist the various techniques of work measurement and explain them in detail.	CO3	6
B) What is the significance of micro motion study in industrial engineering? Give the detail procedure to perform micro motion study.	CO1	6
C) Explain the concept of Just in time in detail and give its advantages.	CO4	6